

Abdullah Ahmed Al-Mekhyal

+965 5580 5805 | a.almekhyal05@gmail.com | Kuwait | [linkedin.com/in/abdullahalmekhyal](https://www.linkedin.com/in/abdullahalmekhyal) | My Portfolio

PROFILE

Computer Science student and Research Assistant at Kuwait University's DATA Lab with experience in AI-driven innovation, digital transformation, and applied technology solutions across finance, education, and operational systems. Experienced in applying machine learning, graph AI, and full-stack development to real-world challenges including Anti-Money Laundering (AML), workforce development, and digital platform transformation. Strong interest in bridging AI, business strategy, and emerging technologies to design scalable solutions, support innovation initiatives, and create measurable organizational impact.

EDUCATION

Kuwait University

Bachelor of Computer Science

Sep 2023 – Present

Kuwait

– Coursework: Database Systems · Machine Learning · Data Mining · Algorithms · NLP · Web Programming · Statistics

PUBLICATIONS

- “**Heterogeneous Graph Modeling for AML: A Reproducible Study on Elliptic++**.” Almekhyal, A. & Joudeh, Y. Kuwait University DATA Lab. ROC-AUC 0.9718, PR-AUC 0.8289, Recall 0.92. github.com/abojoudah/HMPNN-MLD-EllipticDataset
- “**ML-Driven Technique for UAV Threat Detection**.” Al-Mekhyal, A. et al. *IEEE SM 2026*, Paper #1571247459. Accepted (52.7% acceptance rate). **Gold Medal**, GCC Cybersecurity Summit, Qatar 2025.

EXPERIENCE

Kuwait University

Research Assistant — DATA Lab

2026 – Present

Kuwait

- Researching heterogeneous graph and hypergraph ML methods for financial crime detection, including AML, fraud, and suspicious transaction analysis.
- Developing AI-driven approaches targeting scalable and reproducible financial intelligence systems applicable to banks and financial institutions across the GCC.

Global Student Research (GSR) — KFUPM

Local Ambassador

2025 – Present

GCC

- Represent Kuwait University in GCC student research and support regional collaboration in fintech, data science, and applied innovation.

Computer Science Club (CSC) — Kuwait University

Secretary

Sep 2024 – Jun 2025

Kuwait

- Coordinated technical workshops, hackathons, and industry engagement initiatives with organizations including Kuwait Finance House (KFH) and KNPC.
- Supported collaboration between students, academia, and industry partners to promote technology adoption, innovation, and professional development.

Kuwait National Petroleum Company (KNPC)

IT Intern — Training Center

Jul – Aug 2024

Kuwait

- Gained exposure to enterprise IT systems, data governance, compliance documentation, and internal technology operations.

PROJECTS

Heterogeneous Graph AML Detection - Elliptic++ | Machine Learning · Graph Neural Networks · PyTorch Geometric | 2025/2026

- Developed a heterogeneous GNN model on Elliptic++ with 1.47M nodes and 4 edge types for illicit wallet classification; achieved ROC-AUC 0.9718, PR-AUC 0.8289, 92% recall, and $P@R1\% = 1.00$.

Tadreeb - Internship & Workforce Development Platform | MERN Stack · Digital Transformation | 2026

- Designed and developed a full-stack platform centralising internship and training opportunities for Kuwaiti students.
- Focused on transforming student career development through digital profiles, opportunity aggregation, and future AI-driven student-company matching.

Mosque Digital Transformation Platform | SaaS · Product Strategy · Web Platform | 2025/2026

- Designed a digital management platform for mosques and Quran centers featuring role-based dashboards, attendance tracking, student monitoring, and parent engagement.
- Focused on modernising operational workflows and enabling digital transformation for educational and religious insti-

tutions.

Motor Vehicle Collisions Database - NYC | Database Systems · Oracle SQL | 2025/2026

– Designed 3NF relational schema and implemented advanced SQL analytics using window functions, aggregations, and structured reporting queries.

SKILLS & LANGUAGES

Programming	C, C++, Python, SQL, JavaScript
AI & Data	Machine Learning, Heterogeneous GNN, HMPNN, PyTorch Geometric, AML Analytics
Innovation	Digital Transformation, AI Innovation, Product Strategy, Stakeholder Communication, Technical Presentations
Databases	Oracle SQL, ER Modelling, Normalisation (1NF–3NF), NoSQL (MongoDB)
Systems	Linux/Unix, System Programming, Client–Server Architecture
Web	MERN Stack (MongoDB, Express, React, Node.js)
Banking	AML, FATF Frameworks, KYC, Financial Transaction Analysis, Risk Analytics
Languages	Arabic (Native), English (Professional Proficiency)